

# Vikram Kamath Cannanure

POSTDOCTORAL FELLOW - SAARLAND UNIVERSITY · HCI, ICTD, AI, EDUCATION

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I design *Thoughtful AI*: human-centered systems for reflection, care, and trust, built with communities across HCI and ICTD.

- **Research** — 18 full papers at top-tier HCI venues (CHI, CSCW, COMPASS, ICTD); systems, datasets, evaluation frameworks.
- **Impact** — AI systems deployed across the Global South; teacher chatbot in Côte d'Ivoire: 400+ teachers, 10,000+ students.
- **Funding** — \$215,000+ in competitive grants, including a Gates Foundation Grand Challenges award.
- **Recognition** — 491 citations, h-index 12 (Google Scholar, 2026); Best Paper Award at ACM COMPASS.
- **Leadership** — Co-led *HCI Across Borders*, a global HCI community; mentored 25+ students; reviewed 80+ papers.

## Research Experience

### Saarland University

Saarbrücken, Germany

POSTDOCTORAL FELLOW

Oct 2023 – present

- Host: Prof. Ingmar Weber · Leading a research area at the intersection of HCI, ICTD, and generative AI
- Co-supervising PhD and master's students on socially grounded AI projects
- 🤝 Co-PI on a \$100,000 Gates Foundation grant to co-design a chatbot improving women's and youth health in rural India
- Instructor for the graduate seminar "AI and the Global South"
- Key publications: [P13, P12, P14, S4]

## Education

### Carnegie Mellon University

Pittsburgh, USA

PhD, HUMAN-COMPUTER INTERACTION

2017–2023

- Advisors: Prof. Tim Brown, Prof. Amy Ogan · Committee: Chinmay Kulkarni, Mayank Goel, Kentaro Toyama (U. Michigan)
- Dissertation: *DIA: Supporting Teacher Professional Development in Low Infrastructure Settings*
- Led multi-year fieldwork and chatbot deployments for teachers in rural Côte d'Ivoire
- 🏆 Awarded \$100,000 Fellowship from the Center for Machine Learning and Health (CMLH)
- Key publications: [P6, P7, P8, P9, P10, P15]

### Carnegie Mellon University

Pittsburgh, USA

MS, HUMAN-COMPUTER INTERACTION

2017–2021

### Carnegie Mellon University

Pittsburgh, USA

MS, EDUCATIONAL TECHNOLOGY AND APPLIED LEARNING SCIENCE

2014–2015

### National Institute of Technology Karnataka

Karnataka, India

B.TECH.

2007–2011

## Grants and Awards (\$215,000+ total)

|            |                                                                                                             |                 |
|------------|-------------------------------------------------------------------------------------------------------------|-----------------|
| 2026–      | <b>Listen at Scale (Gram Vaani)</b> , Voice-first AI for reproductive and mental wellness — <b>\$5,000</b>  | India           |
| 2025–      | <b>Gates Foundation Grand Challenges (Co-PI)</b> , LLM wellness platform for rural India — <b>\$100,000</b> | India           |
| 2019       | <b>PhD Fellowship</b> , Center for Machine Learning and Health, CMU — <b>\$100,000</b>                      | Pittsburgh, USA |
| 2023, 2025 | <b>SIGCHI Development Grant</b> , HCI Across Borders (HCIXB) workshop at CHI — <b>\$10,000</b>              |                 |
| 2019       | <b>Best Paper Award</b> , ACM COMPASS                                                                       | Accra, Ghana    |
| 2019       | <b>Poster Spotlight</b> , ACM COMPASS                                                                       | Accra, Ghana    |

## Publications

- **Record** — 35+ peer-reviewed publications, including 18 full papers at *CHI*, *COMPASS*, *ICTD*, and *CUI*; 4 manuscripts under review.
- **Metrics (2026)** — 491 citations, h-index 12, i10-index 14. [Google Scholar] [ORCID]
- **Author order** — first = primary intellectual lead; last = supervision and project leadership; \* = equal contribution.

### JOURNAL AND CONFERENCE FULL PAPERS [P]

- [P18] **Cannanure, V.K.**, Yinkfu, B., Bryan, D., Amin, M., Weber, I., *Teacher Professional Development on WhatsApp with LLMs: Early Lessons from Cameroon*. Accepted at Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2026), Springer LNCS. [Link]
- [P17] Behmanush, H., Akhtari, F., Weber, I., **Cannanure, V.K.**, *Designing Safe and Accountable GenAI as a Learning Companion with Women Banned from Formal Education*. Accepted at Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026), ACM. [Link]

- [P16] Wolf, S.\*, Ogan, A.\*, Jasińska, K.\*, Kembou, S.\*, **Cannanure, V.K.**, Whitehead, H., Corbett, J., Tanoh, F., Wortsman, B., Lim, J., Zhao, H., Kumar, A. *Impacts of Targeted Instruction with Mobile-based Support on Teachers and Students in Rural Côte d'Ivoire*. Accepted at American Educational Research Journal. [\[Link\]](#)
- [P15] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T.X., Ogan A. *A Design-Based Research Approach to Bridge Teacher Aspirations and Goal-Setting*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P14] Zhu J., Kempermann M., **Cannanure V.K.**, Hartland A., Navarrete R., Carteny G., Braun D., Weber I. *Learn, Explore and Reflect by Chatting: Understanding the Value of an LLM-Based Voting Advice Application*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P13] Adhikari D.M., Hartland A., Weber I., **Cannanure V.K.**. *Exploring LLMs for Automated Generation and Adaptation of Questionnaires*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P12] Behmanush H., Akhtari F., Nooripoor R., Weber I., **Cannanure V.K.** *Online Learning and GenAI: Supporting Women's Aspirations Amid Socio-Political Instability in Afghanistan*. *ACM COMPASS 2025* [\[Link\]](#)
- [P11] Verma R., Weber I., **Cannanure V.K.**. *Exploring AI in Education: Child-Computer Interaction in the Global South*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P10] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T., Ogan A. *Understanding the Longitudinal Impacts of a Chatbot to Facilitate a Virtual Community of Practice for Teachers in Rural Côte d'Ivoire*. *ACM COMPASS 2024*, Delhi, India. [\[Link\]](#)
- [P9] Ngoon T., **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *"I believe I did not preach into the desert": Opportunities & Challenges in Scaling Teacher Mentorship through Mobile Technology in Rural Côte d'Ivoire*. *L@Scale 2024*, Atlanta, USA. [\[Link\]](#)
- [P8] **Cannanure V.K.**, Ávila E., Ngoon T., Adjii Y., Wolf S., Jasińska K., Brown T., Ogan A. *"We dream of climbing the ladder, for getting there, we have to do our job better": Designing for Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2022*, Seattle, USA. [\[Link\]](#)
- [P7] **Cannanure V.K.**, Brown T., Ogan A. *DIA: A Human-AI Hybrid Chatbot for Low-Resource Contexts*. *ICTD 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P6] **Cannanure V.K.**, Souvenir J., Adjii Y., Apke H., Jasińska K., Brown T., Ogan A. *"I'm fine where I am, but I want to do more": Exploring Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P5] Madaio M., Yarzebinski E., **Cannanure V.K.**, Zinszer B., Seri A., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *Collective Support and Independent Learning with a Voice-Based Literacy Technology in Rural Communities*. *CHI 2020*, Hawaii, USA. [\[Link\]](#)
- [P4]  Madaio M., **Cannanure V.K.**, Yarzebinski E., Zasacky S., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *"You Give a Little of Yourself": Family Support for Children's Use of an IVR Literacy System*. *ACM COMPASS 2019*, Accra, Ghana. (Best Paper Award) [\[Link\]](#)
- [P3] **Cannanure V.K.**, Brown T. *RASP-IVR: A Low-Cost Interactive Voice Response System*. *AfriCHI 2018*, Windhoek, Namibia. [\[Link\]](#)
- [P2] Onafuwa D., Early K., Vyas N., **Cannanure V.K.**, Mankoff J. *Understanding the Information Needs of Prospective Tenants*. *ACM COMPASS 2018*, San Francisco, USA. [\[Link\]](#)
- [P1] Bhat K.\*, Thinayane M.\*, Goldkind L., **Cannanure V.K.** *Critical Participatory Design: Reflections on Engagement and Empowerment in a Case of a Community-Based Organization*. *Participatory Design Conference (PDC 2018)*, Hasselt and Genk, Belgium. [\[Link\]](#)

## BOOK CHAPTERS [B]

- [B1] Gautam A., Wong-Villacrés M., Gaytán Lugo L.S., **Cannanure V.K.**, Vannini S., Jimenez A., Tuli A., Reynolds-Cuellar P., Pei L., Alvarado Garcia A., Badillo-Urquiola K., Prietch S.S., Densmore M., Banganayi T., Dray S. *Human-Computer Interaction Across Borders: Perspectives from the Global South*. In *The Human-Computer Interaction Handbook* (4th ed.), Julie A. Jacko (Ed.). CRC Press, 2026.

## SHORT PAPERS AND POSTERS [S]

- [S5] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *Enabling a Virtual Community of Practice Facilitated by a Chatbot for Rural Teachers*. *Proceedings of the 5th African Human-Computer Interaction Conference (AfriCHI 2025)*, Cairo, Egypt. [\[Link\]](#)

- [S4] Mensah E.E., **Cannanure V.K.**, Weber I. *Ethnic Diversity and Spatial Dynamics in Google Maps Reviews: Insights from a German Border City*. ICWSM 2025 (poster). [\[Link\]](#)
- [S3] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T.X., Ogan A. *Applying Think-Aloud in ICTD: A Case Study of a Chatbot Use by Teachers in Rural Côte d'Ivoire*. ICTD 2024, Nairobi, Kenya. [\[arXiv\]](#)
- [S2] Adhikari D.M., **Cannanure V.K.**, Hartland A., Weber I. *Exploring LLMs for Automated Pre-Testing of Cross-Cultural Surveys*. ICTD 2024, Nairobi, Kenya. [\[arXiv\]](#)
- [S1] **Cannanure V.K.**, Chen X., Mankoff J. *Twist and Knock: A One-Handed Gesture for Smart Watches*. *Proceedings of the 42nd Graphics Interface Conference (GI 2016)*, Victoria, Canada. [\[Link\]](#)

## WORKSHOP PAPERS AND LIGHTLY REVIEWED WORK [L]

- [L4] **Cannanure V.K.**, Varghese D., Rivera-Loaiza C., Noor F., Das D., Jain P., Chang-Rizzo M., Wong-Villacrés M., Karusala N., Ahmed N., Till S., Akhigbe B., Dray S., Densmore M., Sturm C., Kumar N. (2023). *HCI Across Borders: Towards Global Solidarity*. CHI '23, Apr 2023, Hamburg, Germany.
- [L3] **Cannanure V.K.**, Karusala N., Rivera-Loaiza C., Prabhakar A. S., Varanasi R. A., Tuli A., Gamage D., Noor F., Nemer, D., Das D., Dray S., Sturm C., Kumar, N. (2022). *HCI Across Borders: Navigating Shifting Borders at CHI*. CHI '22, Apr 2022, New Orleans, USA.
- [L2] **Cannanure V.K.**, Gamage D., Sturm C., Winschiers-Theophilus H., Maestre, J. F., Karusala N., Reynolds-Cuellar P., Kumar, N., *Workshop: Decolonizing HCI Across Borders*. CHI '21, Apr 2021, Yokohama, Japan.
- [L1] Kumar N., **Cannanure V.K.**, Gamage D., Prabhakar A., Sturm C. *Workshop: HCI Across Borders and Sustainable Development Goals*. CHI '20, Apr 2020, Hawaii, USA.

## Teaching Experience

### AI and the Global South (Block Seminar)

LEAD INSTRUCTOR AND COURSE CREATOR

Student evaluation: 1.22 overall vs. department average 1.51 (lower is better)

Saarland University

Summer 2025

### Elements of Data Science and Artificial Intelligence

COURSE COORDINATOR (INSTRUCTOR: PROF. INGMAR WEBER)

Saarland University

Fall 2024

### Programming Usable Interfaces (05-430/05-630)

TEACHING ASSISTANT, HYBRID (INSTRUCTOR: PROF. SCOTT HUDSON)

Carnegie Mellon University

Spring 2022

### User-Centered Research & Evaluation (05-410/05-610)

TEACHING ASSISTANT, REMOTE (INSTRUCTOR: RAELEN MUSURACA)

Carnegie Mellon University

Spring 2021

## Mentoring

### PhD Students (2)

\***Hamayoon Behmanush**: Saarland University, PhD: *Supporting Afghan women in learning programming* ([P12](#), [P17](#))

\***Ethel Mensah**: Saarland University, PhD: *Exploring structural inequality in Germany using POI data* ([S4](#))

### Master's Students (21)

\***Divya Mani Adhikari**: Saarland University, MS 2025: *Automated Survey Testing using LLMs* ([P13](#), [S2](#))

**Prithika Acharya**: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*

**Aastha Patel**: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*

**Egide Kamuzinzi**: CMU Africa, MSIT 2021: *Teacher Training Project in Côte d'Ivoire*

**Baptiste Moreau-Pernet**: CMU Pittsburgh, METALS 2021: *Teacher Training Project in Côte d'Ivoire*

**Saumya Sharma**: CMU Pittsburgh, MSIIPM 2020: *Simple Health Information System*

**Lucie Uwimpaye**: CMU Africa, MSIT 2021: *Simple Health Information System*

**Eloisa Ávila**: CMU Pittsburgh, MSPPM 2021: *Teacher Training Project in Côte d'Ivoire*

**Sheila Mbadi**: CMU Africa, MSIT 2021: *Simple Health Information System*

**Patrick Manirakiza**: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

**Solange Iyubu**: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

**Sabin Sheja**: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

**Justin Bigwi**: CMU Africa, MSECE 2019: *Design of a Phone Call-Based Health Information System*

**Gloria Umutesi**: CMU Africa, MSECE 2020: *Design of a Phone Call-Based Health Information System*

**Jocelyne Mukamisha**: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*

**Justin Souvenir Niweteto:** CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System* ([P6](#))  
**Fraterne Kagame:** CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*  
**Huguette Ingenere:** CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*  
**Marie Claudine Ingenere:** CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*  
**Nebiyu Yismaw:** CMU Africa, MSECE 2018: *Digital and voice-based systems to support gender-based violence response*  
**Fabiola Uwashema:** CMU Africa, MSIT 2018: *Digital and voice-based systems to support gender-based violence response*

### Undergraduate Students (3)

\***Riti Verma:** Indian Institute of Technology, Delhi: *Exploring the role of AI in the Global South* ([P11](#))  
**Lena Armstrong:** University of Pennsylvania, Summer REU: *Teacher Training Project in Côte d'Ivoire*  
**Maya Greenholt:** CMU Pittsburgh, Cognitive Science: *Teacher Training Project in Côte d'Ivoire*

### Thesis Committees (1)

**Ahana Ghosh:** MPI-SWS, PhD 2026: committee member

\*Mentored during my postdoctoral fellowship at Saarland University.

## Professional Service

### CHAIRING AND ORGANIZING

|           |                                                                            |                  |
|-----------|----------------------------------------------------------------------------|------------------|
| 2026      | <b>Papers Chair</b> , ACM COMPASS 2026                                     | Virtual          |
| 2025      | <b>Lead Mentor</b> , HCI Across Borders (HCIXB) workshop at CHI 2025       | Yokohama, Japan  |
| 2024      | <b>Lead Organizer</b> , HCI Across Borders (HCIXB) workshop at ACM COMPASS | Delhi, India     |
| 2023      | <b>Lead Organizer</b> , HCI Across Borders (HCIXB) workshop at CHI         | Hamburg, Germany |
| 2022      | <b>Lead Organizer</b> , HCI Across Borders (HCIXB) workshop at CHI         | New Orleans, USA |
| 2021      | <b>Lead Organizer</b> , HCI Across Borders (HCIXB) workshop at CHI         | Yokohama, Japan  |
| 2020      | <b>Organizer</b> , HCI Across Borders (HCIXB) symposium at CHI             | Hawaii, USA      |
| 2019      | <b>Organizer</b> , HCI Across Borders (HCIXB) symposium at CHI             | Glasgow, UK      |
| 2019      | <b>Organizer</b> , Young Researchers Reception at ICTD                     | Ahmedabad, India |
| 2020–2022 | <b>Executive Committee Member</b> , SIGCHI Development Fund                |                  |

### PROGRAM COMMITTEES AND REVIEWING

|           |                                                                               |
|-----------|-------------------------------------------------------------------------------|
| 2024–     | <b>Program Committee</b> , ACM COMPASS 2025, AfriCHI 2025, ICTD 2024          |
| 2020–2023 | <b>Associate Chair</b> , CHI Extended Abstracts                               |
| 2018–     | <b>Reviewer (80+ papers)</b> , CHI, CSCW, ACM COMPASS, ICTD, CHI Case Studies |

## Professional Experience

### Carnegie Mellon University, Africa

VISITING RESEARCHER

Kigali, Rwanda  
2018–2020 (summers, spring)

- Host: Prof. Tim Brown
- Project: Design of a Phone Call-Based Health Information System
- Conducted participatory research with rural women and prototyped an IVR system in rural Rwanda

### United Nations University Computing and Society

RESEARCH INTERN

Macau, SAR, China  
Jul 2017 – Sep 2017

- Advisor: Dr. Mamello Thinyane
- Conducted participatory design with a local NGO
- Built a mobile app to report and monitor homelessness in Macau

### Carnegie Mellon University

SENIOR WEB DEVELOPER

Pittsburgh, USA  
Jul 2015 – Jun 2017

- Advisor: Prof. Geoff Gordon
- Built and managed educational technology to support online learning
- Prototyped a low-cost video recording system for teaching courses

### Carnegie Mellon University

RESEARCH ASSISTANT

Pittsburgh, USA  
Aug 2015 – Jun 2017

- Advisor: Prof. Jen Mankoff
- Longitudinal eating detection: deployed and evaluated the system with 12 users in the field
- Twist & Knock: a one-handed gesture for smartwatches ([S1](#))
- EDigs: built and maintained the APIs for an apartment-hunting website in Pittsburgh

- Engineered APIs for Android, iOS, web, and Google Glass
- Scaled back-end to handle 4 million users and manage 100K locations
- Designed a 3D viewer on par with Google Street View
- Managed 10 Ubuntu machines with MySQL and Redis replication

## Skills

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|                            |                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <b>Methods</b>             | Participatory design, longitudinal field deployment, interviews, think-aloud, mixed-methods analysis |
| <b>Programming</b>         | Python, JavaScript, PHP, HTML/CSS, Android, iOS, Arduino                                             |
| <b>Tools and Platforms</b> | LLM APIs, chatbot and IVR platforms, WhatsApp deployment, pandas, scikit-learn, MySQL, Redis         |

## Selected Press Coverage

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**The Next Web** — *TeliportMe introduces 3D effects to its 360-degree panoramic app for Android*

**TechCrunch** — *Test Flighting: Applover Launches Free, Crowdsourced Testing Platform for Android Apps*

**Robert Scoble** — *Developers: crowd-source your testing of Android phones*

**CMU** — *A new online resource from Carnegie Mellon University seeks to help people find the perfect rental*

**CMU HCII** — *EDigs Helps Apartment Hunters Find the Perfect Rental*

**The Better India** — *How a National Level Cyclist Is Fighting Both Spinal Injury & Poverty with Inspiring Confidence*

**Milaap Blog** — *Getting Back to Cycling: A National Medalist's Dream*

## Beyond Research

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- **Philanthropy** — Raised over **\$20,000** for causes in India through swimming and crowdfunding
- **Android** — Shipped 3 apps to the Play Store, with roughly 10,000 active users
- **Web** — Deployed 20+ websites to production; the largest reached over **4 million users**
- **Teaching** — Taught Kannada as a second language to children at the S.V. Temple in Pittsburgh
- **Swimming** — Swam 10 miles from Elephanta Caves to the Gateway of India to raise money for acid-attack survivors